Structural Inductive Bias Reduces Sample Complexity in Implicit Neural Models for Dynamical Systems

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Abstract

We prove that implicit neural models with structural inductive bias achieve tighter generalization bounds than explicit models, and therefore require fewer training samples for the same approximation quality. The key mechanism is that an implicit formulation $F(x, y, \mathrm{MLP}(x; \theta)) = 0$, where F encodes known structural knowledge about the target function, *changes* the learning target: the network approximates a simpler residual h^* instead of the original function f^*. Using Rademacher complexity theory, we derive explicit generalization bounds for both explicit and implicit formulations, and establish a quantitative gain condition under which the implicit model achieves lower sample complexity. We define the *structural content* of F as the ratio of path norm reduction, and show that the implicit model outperforms the explicit one if and only if the structural content exceeds an amplification factor determined by the Lipschitz constant of F. We provide sufficient conditions for complexity reduction (structural factorization, nonlinearity absorption), counterexamples showing when implicit models fail to help, and two instantiations of the framework: physics-informed variable transformations, where F absorbs known solution structure such as oscillatory or exponential behavior, reducing the learning target to a slowly varying modulation; and the Homotopy Analysis Method (HAM), where F encodes the governing differential equation and decomposes the solution into a series of linear corrections of decreasing complexity. We distinguish this target-transformation mechanism from the regularization approach of Physics-Informed Neural Networks (PINNs), which constrains the optimization landscape but does not change the learning target. The connection to a deeper topological framework based on autosimilar function spaces is identified as a direction for future work.

Keywords: Sample complexity; Rademacher complexity; structural inductive bias; implicit neural models; Homotopy Analysis Method; physics-informed architectures

1 Introduction

Neural networks have become the dominant tool for approximating unknown functions from data, with universal approximation theorems guaranteeing that sufficiently large networks can

approximate any continuous function on a compact domain [1, 2, 3]. However, universality is a statement about *capacity*, not about *efficiency*: knowing that a network *can* approximate f^* says nothing about how many parameters or training samples are needed to do so reliably. The gap between capacity and efficiency is governed by the *complexity* of the hypothesis class relative to the target function.

In the context of dynamical systems, the target function f^* is not arbitrary: it is constrained by physical laws, symmetries, and structural regularities. A neural network that ignores these constraints must discover all structure from data alone. A network that exploits them — by embedding known structure into the model — should, in principle, require less data for the same approximation quality. This intuition is widely held but rarely formalized.

Several strategies have been proposed for injecting physical knowledge into neural models. Physics-Informed Neural Networks (PINNs) [9] add a PDE residual term to the loss function, penalizing solutions incompatible with the governing equation. Neural ODEs [10] parametrize the vector field $\dot{x} = f_\theta(x)$ and integrate it with a standard solver. Deep Equilibrium Models (DEQs) [11] define the output as a fixed point of an implicit layer. These approaches constrain the optimization landscape but share a common feature: the network still approximates the full target function f^* directly. The physics acts as a regularizer, not as a simplifier of the learning target.

A fundamentally different strategy is to *transform* the learning target by encoding structural knowledge in the model architecture. If $f^*(x) = g(x) \cdot e^{-\lambda x}$ with λ known, one can define $\hat{f}(x) = \mathrm{MLP}(x;\theta) \cdot e^{-\lambda x}$, so the network learns only the slowly varying modulation g instead of the full product f^*. The known exponential structure is *absorbed* by the model; the network's learning target is genuinely simpler.

The present paper formalizes this target-transformation mechanism. We prove that when a known function F encodes structural knowledge about f^* and the network participates in an implicit equation $F(x, y, \mathrm{MLP}(x;\theta)) = 0$, the learning target changes from f^* to a simpler residual h^*, and the sample complexity decreases quantifiably.

Remark 1 (Distinction from PINNs)**.** *It is important to distinguish the mechanism formalized here from the PINN approach [9]. In a PINN, the network $u_\theta(x)$ approximates f^* directly; the PDE residual $\mathcal{N}[u_\theta]$ is added to the loss as a regularizer. The learning target remains f^* — the network must represent the full solution. In our framework, the implicit function F* changes *the learning target from f^* to a different, simpler function h^* defined by $F(x, f^*(x), h^*(x)) = 0$. The network approximates h^*, which has lower complexity.*

PINNs reduce effective complexity by constraining the optimization landscape (a property of the training algorithm). The present framework reduces effective complexity by reducing the intrinsic difficulty of the approximation target *(a property of the function class), which translates into tighter generalization bounds independently of the optimization method. The two mechanisms are complementary, not competing: one could use a PINN-style residual loss* in addition to *a target-transforming implicit model.*

1.1 Contributions

1. We derive explicit Rademacher complexity bounds for both explicit and implicit formulations (Theorem 6), with a rigorous treatment of the Lipschitz composition and

centering correction required by the implicit model.

2. We establish a quantitative *gain condition* (Corollary 7) under which the implicit model requires fewer training samples, expressed in terms of the path norm reduction and the Lipschitz constant of F.

3. We define the *structural content* of F (Definition 8) as the ratio of path norm bounds, and provide sufficient conditions for F to have positive structural content (Propositions 9 and 10).

4. We show that physics-informed variable transformations and HAM-based decompositions are instances of this framework (Section 9), explaining why architectures such as SIREN [12] and Laguerre networks outperform generic MLPs for specific dynamical classes.

5. We provide counterexamples (Section 8) showing that not every implicit formulation reduces complexity, and characterize the conditions under which the implicit model is actually worse.

The connection between the structural inductive bias formalized here and a deeper topological framework based on autosimilar function spaces and iterated function systems is identified as a direction for future work (Section 11).

2 Problem Formulation

Definition 1 (Learning scenario)**.** *Let $\mathcal{X} \subset \mathbb{R}^n$ be compact with $\|x\| \leq X_{\max}$ for all $x \in \mathcal{X}$. Let $f^* : \mathcal{X} \to \mathbb{R}$ be the unknown target function. A training set $S = \{(x_i, y_i)\}_{i=1}^N$ is available with $y_i = f^*(x_i) + \varepsilon_i$, where ε_i is sub-Gaussian noise with parameter σ^2.*

Definition 2 (Hypothesis class)**.** *For an MLP of depth L and maximum width d, the hypothesis class with bounded spectral path norm is:*

$$\mathcal{H}_B = \left\{ \mathrm{MLP}(\cdot\,;\theta) : \prod_{l=1}^{L} \|W_l\|_\sigma \leq B \right\} \tag{1}$$

where $\|W_l\|_\sigma$ is the spectral norm of the weight matrix at layer l and $B > 0$ is the path norm bound.

We consider two formulations for learning f^*:

Explicit formulation. The network approximates f^* directly:

$$\hat{\theta}_{\mathrm{exp}} = \arg\min_\theta \ \hat{\mathcal{L}}_{\mathrm{exp}}(\theta) = \arg\min_\theta \ \frac{1}{N} \sum_{i=1}^{N} (y_i - \mathrm{MLP}(x_i;\theta))^2. \tag{2}$$

Implicit formulation. Let $F : \mathcal{X} \times \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ be a known function of class C^1. The network participates as a component of an implicit equation:

$$\hat{\theta}_{\mathrm{imp}} = \arg\min_\theta \ \hat{\mathcal{L}}_{\mathrm{imp}}(\theta) = \arg\min_\theta \ \frac{1}{N} \sum_{i=1}^{N} F(x_i, y_i, \mathrm{MLP}(x_i;\theta))^2. \tag{3}$$

The function F encodes structural knowledge about the relationship between x and y. When $F(x, y, z) = y - z$, the implicit formulation reduces to the explicit one. When F is nontrivial, it absorbs part of the complexity of f^*, so that the network learns a different, simpler function $h^* \neq f^*$.

3 Formal Hypotheses

Assumption 1 (Regularity of F)**.** *The function $F : \mathcal{X} \times \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ satisfies:*

(a) F is C^1 in all variables.

(b) ***Local Lipschitz in the training region.*** *Let $\mathcal{R} = \mathcal{X} \times \mathcal{Y} \times \mathcal{Z}$ be the training region, where $\mathcal{Y} \supset \{f^*(x) : x \in \mathcal{X}\}$ and $\mathcal{Z} \supset \{h^*(x) : x \in \mathcal{X}\}$ are compact neighborhoods. There exist constants $0 < \alpha \leq \beta$ such that:*

$$\alpha \;\leq\; \left|\frac{\partial F}{\partial z}(x, y, z)\right| \;\leq\; \beta \qquad \forall\,(x, y, z) \in \mathcal{R}. \tag{4}$$

The constant β is local*: it depends on the size of $\mathcal{R}$ and is not required to hold globally.*

(c) $\frac{\partial F}{\partial y}(x, y, z) \neq 0$ on $\mathcal{R}$ (Implicit Function Theorem condition).

Assumption 2 (Existence of implicit target)**.** *Under Assumption 1, there exists a function $h^* : \mathcal{X} \to \mathbb{R}$ of class C^1, defined implicitly by:*

$$F(x, f^*(x), h^*(x)) = 0 \qquad \forall\, x \in \mathcal{X}. \tag{5}$$

The function h^* is what the MLP must learn in the implicit formulation.

Assumption 3 (Complexity reduction)**.** *There exist path norm bounds B_{exp} and B_{imp} such that:*

(a) f^* is ε_0-approximable by $\mathcal{H}_{B_{\text{exp}}}$.

(b) h^* is ε_0-approximable by $\mathcal{H}_{B_{\text{imp}}}$.

(c) $B_{\text{imp}} < B_{\text{exp}}$.

Remark 2 (Nature of Assumption 3)**.** *Assumption 3 is the substantive hypothesis. Assumptions 1 and 2 are technical regularity conditions. The claim "the implicit model requires fewer data" is exactly as strong as Assumption 3 — it depends entirely on how much complexity F absorbs. Sections 7 and 8 provide conditions under which this assumption holds or fails.*

4 Preliminary Results

Lemma 3 (Rademacher complexity of norm-bounded networks — Golowich, Rakhlin & Shamir [6])**.** *For the class $\mathcal{H}_B$ of networks of depth L, maximum width d, and spectral path norm bounded by B:*

$$\mathfrak{R}_N(\mathcal{H}_B) \;\leq\; \frac{B \cdot X_{\max} \cdot \sqrt{2L\log(2d)}}{\sqrt{N}}. \tag{6}$$

Remark 3 (On the simplification)**.** *The full bound of Golowich et al. involves per-layer spectral norms and $(2,1)$-norms:*

$$\mathfrak{R}_N(\mathcal{H}_B) \;\leq\; \frac{\left(\prod_{l=1}^{L} \|W_l\|_\sigma\right) \cdot X_{\max} \cdot \sqrt{\sum_{l=1}^{L} \left(\|W_l\|_{2,1} \,/\, \|W_l\|_\sigma\right)^{2/3}}}{\sqrt{N}}. \tag{7}$$

Bound (6) follows from $\|W_l\|_{2,1} \,/\, \|W_l\|_\sigma \leq \sqrt{d}$. This preserves the asymptotic order $O(1/\sqrt{N})$ but may be loose for structurally sparse weight matrices.

Lemma 4 (Contraction lemma — Ledoux & Talagrand [7])**.** *Let $\phi : \mathbb{R} \to \mathbb{R}$ be L_ϕ-Lipschitz with $\phi(0) = 0$, and let $\mathcal{G}$ be a function class. Then:*

$$\mathfrak{R}_N(\phi \circ \mathcal{G}) \;\leq\; L_\phi \cdot \mathfrak{R}_N(\mathcal{G}). \tag{8}$$

Lemma 5 (Uniform generalization bound — Bartlett & Mendelson [5])**.** *Let $\mathcal{G}$ be a function class with values in $[0, M]$. Then, with probability at least $1-\delta$ over S:*

$$\sup_{g \in \mathcal{G}} \left| \mathbb{E}[g] - \hat{\mathbb{E}}_N[g] \right| \;\leq\; 2\mathfrak{R}_N(\mathcal{G}) + 3M\sqrt{\frac{\log(2/\delta)}{2N}}. \tag{9}$$

5 Main Theorem

Theorem 6 (Generalization bound reduction by implicit model)**.** *Under Assumptions 1–3, let $\hat{\theta}_{\text{exp}} \in \mathcal{H}_{B_{\text{exp}}}$ and $\hat{\theta}_{\text{imp}} \in \mathcal{H}_{B_{\text{imp}}}$ be the empirical risk minimizers of $\hat{\mathcal{L}}_{\text{exp}}$ and $\hat{\mathcal{L}}_{\text{imp}}$ respectively. Then, with probability at least $1-\delta$:*

(I) Explicit bound:

$$\mathcal{L}_{\text{exp}}(\hat{\theta}_{\text{exp}}) \;\leq\; \hat{\mathcal{L}}_{\text{exp}}(\hat{\theta}_{\text{exp}}) + \frac{4M_{\text{exp}} \cdot B_{\text{exp}} \cdot X_{\max} \cdot \sqrt{2L\log(2d)}}{\sqrt{N}} + 3M_{\text{exp}}^2\sqrt{\frac{\log(2/\delta)}{2N}} \tag{10}$$

(II) Implicit bound:

$$\mathcal{L}_{\text{imp}}(\hat{\theta}_{\text{imp}}) \;\leq\; \hat{\mathcal{L}}_{\text{imp}}(\hat{\theta}_{\text{imp}}) + \frac{4\beta \cdot M_{\text{imp}} \cdot B_{\text{imp}} \cdot X_{\max} \cdot \sqrt{2L\log(2d)}}{\sqrt{N}} + 3M_{\text{imp}}^2\sqrt{\frac{\log(2/\delta)}{2N}} \tag{11}$$

where $M_{\text{exp}} = \sup_{(x,\theta)} |y - \text{MLP}(x;\theta)|$ and $M_{\text{imp}} = \sup_{(x,\theta)} |F(x, y, \text{MLP}(x;\theta))|$ are range bounds over $\mathcal{H}_{B_{\text{exp}}}$ and $\mathcal{H}_{B_{\text{imp}}}$ respectively.

Proof. **Part (I): Explicit bound.**

Define the loss class:

$$\mathcal{G}_{\text{exp}} = \{(x, y) \mapsto (y - g(x))^2 : g \in \mathcal{H}_{B_{\text{exp}}}\}. \tag{12}$$

Step 1: Lipschitz composition. The function $\phi(t) = t^2$ is $2M_{\text{exp}}$-Lipschitz on $[-M_{\text{exp}}, M_{\text{exp}}]$. The inner class $\tilde{\mathcal{G}}_{\text{exp}} = \{(x, y) \mapsto y - g(x) : g \in \mathcal{H}_{B_{\text{exp}}}\}$ satisfies $\mathfrak{R}_N(\tilde{\mathcal{G}}_{\text{exp}}) = \mathfrak{R}_N(\mathcal{H}_{B_{\text{exp}}})$ (translation by y does not affect Rademacher complexity, since y is fixed with respect to the hypothesis class).

To apply the contraction lemma (Lemma 4), we note that $\phi(t) = t^2$ does not satisfy $\phi(0) = 0$. However, the Rademacher complexity is invariant under constant shifts of the function class: for any constant c, $\mathfrak{R}_N(\{g + c : g \in \mathcal{G}\}) = \mathfrak{R}_N(\mathcal{G})$. Since $(y - g(x))^2 = \psi(y - g(x))$ where $\psi(t) = t^2$ and $\psi(0) = 0$, the contraction lemma applies to the centered class $\tilde{\mathcal{G}}_{\text{exp}}$ with Lipschitz constant $2M_{\text{exp}}$. Therefore:

$$\mathfrak{R}_N(\mathcal{G}_{\text{exp}}) \;\leq\; 2M_{\text{exp}} \cdot \mathfrak{R}_N(\mathcal{H}_{B_{\text{exp}}}). \tag{13}$$

Step 2: Rademacher substitution. By Lemma 3:

$$\mathfrak{R}_N(\mathcal{G}_{\text{exp}}) \;\leq\; \frac{2M_{\text{exp}} \cdot B_{\text{exp}} \cdot X_{\max} \cdot \sqrt{2L \log(2d)}}{\sqrt{N}}. \tag{14}$$

Step 3: Generalization bound. Applying Lemma 5 yields (10). $\square_{\text{I}}$

Part (II): Implicit bound.

Define:

$$\mathcal{G}_{\text{imp}} = \{(x, y) \mapsto F(x, y, g(x))^2 : g \in \mathcal{H}_{B_{\text{imp}}}\}. \tag{15}$$

Step 1: Lipschitz composition with centering correction. The composition has two layers:

(i) $z \mapsto F(x, y, z)$ is β-Lipschitz in z by Assumption 1(b).

(ii) $t \mapsto t^2$ is $2M_{\text{imp}}$-Lipschitz on $[-M_{\text{imp}}, M_{\text{imp}}]$.

The contraction lemma (Lemma 4) requires the Lipschitz map to satisfy $\phi(0) = 0$. The map $z \mapsto F(x, y, z)$ does not satisfy $F(x, y, 0) = 0$ in general. We handle this by defining the centered map $\tilde{F}_{x,y}(z) := F(x, y, z) - F(x, y, 0)$, which satisfies $\tilde{F}_{x,y}(0) = 0$ and is β-Lipschitz (same constant). Since Rademacher complexity is invariant under addition of a function that depends only on the data (x, y) and not on the hypothesis g:

$$\mathfrak{R}_N\big(\{(x, y) \mapsto F(x, y, g(x)) : g \in \mathcal{H}_{B_{\text{imp}}}\}\big) = \mathfrak{R}_N\big(\{(x, y) \mapsto \tilde{F}_{x,y}(g(x)) : g \in \mathcal{H}_{B_{\text{imp}}}\}\big). \tag{16}$$

We now apply the contraction lemma twice. First, to $\tilde{F}_{x,y}$ (which is β-Lipschitz with $\tilde{F}_{x,y}(0) = 0$):

$$\mathfrak{R}_N\big(\{\tilde{F}_{x,y} \circ g : g \in \mathcal{H}_{B_{\text{imp}}}\}\big) \;\leq\; \beta \cdot \mathfrak{R}_N(\mathcal{H}_{B_{\text{imp}}}). \tag{17}$$

Second, the squaring map $\psi(t) = t^2$ satisfies $\psi(0) = 0$ and is $2M_{\text{imp}}$-Lipschitz on the relevant range. Therefore:

$$\mathfrak{R}_N(\mathcal{G}_{\text{imp}}) \;\leq\; 2M_{\text{imp}} \cdot \beta \cdot \mathfrak{R}_N(\mathcal{H}_{B_{\text{imp}}}). \tag{18}$$

Step 2: Rademacher substitution. By Lemma 3:

$$\mathfrak{R}_N(\mathcal{G}_{\text{imp}}) \;\leq\; \frac{2\beta \cdot M_{\text{imp}} \cdot B_{\text{imp}} \cdot X_{\max} \cdot \sqrt{2L\log(2d)}}{\sqrt{N}}. \tag{19}$$

Step 3: Generalization bound. Applying Lemma 5 yields (11). $\square_{\text{II}}$ $\square$

Remark 4 (Role of β)**.** *The Lipschitz constant β of F in z appears as an amplification factor in the implicit bound* (11)*. This is not an artifact of the proof: it reflects the genuine cost of passing through the nonlinear map F. If F is "rigid" (β large), the benefit of the reduced path norm $B_{\text{imp}} < B_{\text{exp}}$ may be offset by the amplification. The gain condition is made precise in Corollary 7.*

6 Sample Complexity Comparison

Corollary 7 (Sample complexity reduction)**.** *Fix a generalization gap $\varepsilon > 0$ and confidence $1-\delta$. The number of training samples required to achieve the generalization gap ε is, up to logarithmic factors:*

Explicit:

$$N_{\text{exp}} = O\left(\frac{M_{\text{exp}}^2 \cdot B_{\text{exp}}^2 \cdot X_{\max}^2 \cdot L\log(d)}{\varepsilon^2}\right). \tag{20}$$

Implicit:

$$N_{\text{imp}} = O\left(\frac{\beta^2 \cdot M_{\text{imp}}^2 \cdot B_{\text{imp}}^2 \cdot X_{\max}^2 \cdot L\log(d)}{\varepsilon^2}\right). \tag{21}$$

The sample complexity ratio is:

$$\boxed{\frac{N_{\text{imp}}}{N_{\text{exp}}} = \frac{\beta^2 \cdot M_{\text{imp}}^2 \cdot B_{\text{imp}}^2}{M_{\text{exp}}^2 \cdot B_{\text{exp}}^2}} \tag{22}$$

Proof. Setting the dominant term in (10) equal to ε and solving for N yields (20). The procedure is identical for (21), with the additional factor β from the Lipschitz constant of F. $\square$

Remark 5 (Gain condition)**.** *The implicit model achieves lower sample complexity when:*

$$\beta^2 \cdot M_{\text{imp}}^2 \cdot B_{\text{imp}}^2 \;<\; M_{\text{exp}}^2 \cdot B_{\text{exp}}^2. \tag{23}$$

Definition 8 (Structural content of F)**.** *The structural content of F is:*

$$\mathcal{S}(F) \;:=\; \frac{B_{\text{exp}}}{B_{\text{imp}}}, \tag{24}$$

with $\mathcal{S}(F) > 1$ indicating that F provides useful structural bias, and $\mathcal{S}(F) = 1$ indicating that F is structurally empty (cf. Counterexample 1). The gain condition (23) *becomes:*

$$\mathcal{S}(F) \;>\; \frac{\beta \cdot M_{\text{imp}}}{M_{\text{exp}}}. \tag{25}$$

Remark 6 (Structural content is problem-dependent)**.** *The structural content $\mathcal{S}(F)$ depends on the pair (f^*, F), not on F alone, since B_{exp} is determined by the unknown f^*. In practice, $\mathcal{S}(F)$ is not computable a priori. However, the sufficient conditions of Section 7 provide classes of problems for which $\mathcal{S}(F) > 1$ can be guaranteed from structural properties of F alone.*

7 Sufficient Conditions for Complexity Reduction

Assumption 3 is the key piece. Under what conditions on F is $B_{\text{imp}} < B_{\text{exp}}$ guaranteed?

Proposition 9 (Reduction by structural factorization)**.** *Suppose F has the form:*

$$F(x, y, z) = \psi(x, y) - z, \tag{26}$$

where $\psi : \mathcal{X} \times \mathbb{R} \to \mathbb{R}$ is a known function. Then $h^*(x) = \psi(x, f^*(x))$, and the Lipschitz constant is $\beta = 1$ (no amplification).

If ψ simplifies the structure of f^* — for example, if $f^*(x) = p(x) \cdot s(x)$ with p a known polynomial and s a smooth function of low complexity, and $\psi(x, y) = y/p(x)$ — then $h^*(x) = s(x)$ and $B_{\text{imp}} \ll B_{\text{exp}}$.

Proof. If $F(x, y, z) = \psi(x, y) - z$, then $F = 0$ implies $z = \psi(x, y)$. Evaluating at $y = f^*(x)$: $h^*(x) = \psi(x, f^*(x))$. The Lipschitz constant is $|\partial F/\partial z| = |-1| = 1$, so $\alpha = \beta = 1$.

For the path norm reduction: let $f^*(x) = p(x) \cdot s(x)$ where p is a known polynomial of degree k and s is smooth with slow variation. If $\psi(x, y) = y/p(x)$, then $h^*(x) = s(x)$. By the Barron norm characterization of network approximation complexity [4], the path norm required to approximate a function is proportional to its spectral variation:

$$B \;\propto\; C_f \;=\; \int_{\mathbb{R}^n} \|\omega\| \, |\hat{f}(\omega)| \, d\omega. \tag{27}$$

The product $f^* = p \cdot s$ corresponds to spectral convolution $\widehat{p \cdot s} = \hat{p} * \hat{s}$, which redistributes the spectral content of s toward higher frequencies when p is nonconstant. For polynomials of degree $k \geq 1$, the spectral measure $\hat{p}$ involves derivatives of the Dirac delta, ensuring redistribution away from the origin. Therefore $C_{f^*} = C_{p \cdot s} \geq C_s = C_{h^*}$ under generic conditions on p, and $B_{\text{imp}} \leq B_{\text{exp}}$. □

Proposition 10 (Reduction by nonlinearity absorption)**.** *Let $f^*(x) = \phi(g(x))$ where ϕ is a known invertible C^1 nonlinearity and g is unknown. Define:*

$$F(x, y, z) = \phi(z) - y. \tag{28}$$

Then $h^*(x) = g(x) = \phi^{-1}(f^*(x))$, and the MLP only needs to learn g instead of $\phi \circ g$.

The Lipschitz constant is $\beta = \sup_{\mathcal{Z}} |\phi'(z)|$, which may be large. The gain condition (23) is satisfied when the reduction in B dominates the amplification by β.

Proof. $F(x, y, z) = \phi(z) - y = 0$ implies $z = \phi^{-1}(y)$. At $y = f^*(x) = \phi(g(x))$: $h^*(x) = \phi^{-1}(\phi(g(x))) = g(x)$.

For $\phi(t) = e^t$: $f^* = e^g$ and $h^* = g = \ln f^*$. The Barron norm of e^g grows exponentially with that of g (via the Taylor expansion of the exponential and iterated spectral convolution), so $C_{f^*} = C_{e^g} \gg C_g = C_{h^*}$.

The Lipschitz constant is $\beta = \sup |e^z| = e^{\sup |g|}$, which can be large. The gain condition is met when the path norm reduction dominates: $B_{\text{exp}}/B_{\text{imp}} > \beta \cdot M_{\text{imp}}/M_{\text{exp}}$. □

8 Counterexamples: When the Implicit Model Fails

Counterexample 1 (F without structural content)**.** *Let $F(x, y, z) = y - z$. Then $h^*(x) = f^*(x)$ and $B_{\text{imp}} = B_{\text{exp}}$. The implicit equation absorbs no complexity; the network must learn exactly the same function. The Lipschitz constant is $\beta = 1$, so there is no amplification, but also no gain: $\mathcal{S}(F) = 1$.*

Counterexample 2 (F that amplifies complexity)**.** *Let $F(x, y, z) = y - e^z$. Then $h^*(x) = \ln(f^*(x))$, which may have singularities if f^* vanishes, and the condition $\alpha > 0$ of Assumption 1 fails in regions where $e^z \to 0$. Even when $f^* > 0$ everywhere, the Lipschitz constant $\beta = \sup |e^z|$ can be large in the training region, potentially making the implicit formulation worse than the explicit one.*

Remark 7 (Diagnostic criterion)**.** *Not every implicit model reduces complexity. The structural content $\mathcal{S}(F)$ is the diagnostic: $\mathcal{S}(F) > 1$ is necessary, and the full gain condition* (25) *is sufficient. Counterexample 1 shows $\mathcal{S} = 1$ (neutral), and Counterexample 2 shows how the amplification factor β can dominate even when $\mathcal{S} > 1$.*

9 Instantiations of the Framework

We show that two strategies for injecting structural knowledge into neural models are genuine instances of the general framework of Theorem 6: both change the learning target from f^* to a simpler function h^*.

9.1 Physics-informed variable transformations

Consider a dynamical system whose solutions are known to exhibit a specific qualitative structure: oscillatory behavior, exponential decay, or polynomial modulation. In each case, the practitioner possesses structural knowledge about f^* that can be absorbed into F, genuinely simplifying the learning target.

Case 1: Exponential envelope absorption. If the solution is known to have the form $f^*(x) = g(x) \cdot e^{-\lambda x}$ where $\lambda > 0$ is known (e.g., a dissipative system with known damping rate) and g is an unknown modulation, define:

$$F(x, y, z) = y \cdot e^{\lambda x} - z. \tag{29}$$

Then $h^*(x) = g(x)$ and $\beta = 1$. The network learns only the slowly varying modulation g instead of the oscillatory, decaying product $f^* = g \cdot e^{-\lambda x}$. Since the exponential envelope

concentrates spectral energy at high frequencies, removing it reduces the Barron norm: $C_{h^*} = C_g \ll C_{g \cdot e^{-\lambda x}} = C_{f^*}$.

This explains why Laguerre-type networks (using $t^n e^{-\lambda t}$ as activation functions) outperform generic MLPs for damped systems: the activation absorbs the exponential structure, and the weights modulate a simpler residual.

Case 2: Polynomial factor extraction. If $f^*(x) = p(x) \cdot s(x)$ where p is a known polynomial (e.g., from a Taylor expansion of the nonlinearity at a known equilibrium) and s is unknown but smooth, define:

$$F(x, y, z) = y/p(x) - z. \tag{30}$$

Then $h^*(x) = s(x)$ and $\beta = 1$. This is a direct instance of Proposition 9.

Case 3: Frequency demodulation. If the solution is known to oscillate at a dominant frequency ω (e.g., an AC circuit, a mechanical resonator), the signal can be decomposed as $f^*(x) = A(x)\cos(\omega x) + B(x)\sin(\omega x)$ where A, B are slowly varying envelopes. Defining two implicit models:

$$F_c(x, y, z) = \langle y, \cos(\omega \cdot)\rangle_{\text{local}} - z, \tag{31}$$

$$F_s(x, y, z) = \langle y, \sin(\omega \cdot)\rangle_{\text{local}} - z, \tag{32}$$

where $\langle \cdot, \cdot \rangle_{\text{local}}$ denotes a windowed inner product, the learning targets become A and B respectively, each with $\beta = 1$.

This is the mechanism underlying SIREN [12]: the sinusoidal activation encodes the carrier frequency, and the network weights learn the slowly varying modulation. The framework explains why SIREN outperforms generic MLPs for wave-like dynamics: the activation absorbs the dominant oscillatory structure (reducing B_{imp}) without amplification ($\beta = 1$).

Remark 8 (Common structure of the three cases)**.** *All three cases share the form $F(x, y, z) = \psi(x, y) - z$ with $\beta = 1$. The known structure (exponential envelope, polynomial factor, carrier frequency) is absorbed by ψ, reducing the learning target to a simpler modulation. The structural content is $\mathcal{S}(F) = B_{\text{exp}}/B_{\text{imp}} > 1$, and the gain condition* (25) *reduces to $\mathcal{S}(F) > M_{\text{imp}}/M_{\text{exp}}$, which is satisfied whenever the modulation has comparable range to the original signal.*

9.2 Homotopy Analysis Method

The Homotopy Analysis Method (HAM) of Liao [13, 14] provides a different and deeper instance of the framework. Rather than absorbing a single known structural feature (an exponential, a polynomial, a frequency), the HAM systematically decomposes the *entire* solution of a nonlinear ODE $\mathcal{N}[u] = 0$ into a series of linear corrections.

The deformation is governed by the homotopy equation:

$$(1 - q)\,\mathcal{L}[u(t; q) - u_0(t)] = q\,\hbar\,H(t)\,\mathcal{N}[u(t; q)], \tag{33}$$

where $\mathcal{L}$ is a chosen auxiliary linear operator, $\hbar \neq 0$ is a convergence-control parameter, $H(t)$ is an auxiliary function, and $q \in [0, 1]$ is the homotopy parameter. Setting $q = 0$ gives $u(t; 0) = u_0(t)$; setting $q = 1$ gives $\mathcal{N}[u(t; 1)] = 0$, the original equation.

The solution expands as:

$$u(t;q) = u_0(t) + \sum_{k=1}^{\infty} u_k(t)\, q^k, \tag{34}$$

where each u_k satisfies a *linear* equation with forcing terms depending on $u_0, \ldots, u_{k-1}$.

In the language of the present framework: the target function f^* is the solution of $\mathcal{N}[u] = 0$ (potentially highly complex); the function h^* that the MLP must learn is the residual correction beyond the K-th order HAM approximation: $h^*(t) = u(t;1) - \sum_{k=0}^{K-1} u_k(t)$. Since the K leading terms are computed analytically by solving linear equations, only the residual h^* requires neural approximation.

Proposition 11 (HAM complexity reduction — conditional)**.** *If the HAM series converges geometrically with ratio $\rho < 1$ (i.e., $\|u_k\|_\infty \le C\rho^k$), and the Barron norm of $h_K^* := u^* - \sum_{k=0}^{K-1} u_k$ satisfies $C_{h_K^*} \le C'\rho^K$ for some constant C', then the path norm required to approximate h_K^* satisfies:*

$$B_{\text{imp}} = O(\rho^K) \ll B_{\text{exp}} \tag{35}$$

for K sufficiently large, and the sample complexity ratio decreases exponentially with the truncation order:

$$\frac{N_{\text{imp}}}{N_{\text{exp}}} = O(\rho^{2K}). \tag{36}$$

Remark 9 (Scope of Proposition 11)**.** *The condition $C_{h_K^*} \le C'\rho^K$ is an additional regularity assumption beyond $\|u_k\|_\infty \le C\rho^k$. The supremum norm and the Barron norm are not equivalent: a function with small $\|\cdot\|_\infty$ may have high spectral variation (e.g., small-amplitude but highly oscillatory functions). However, for the HAM residuals h_K^*, which are solutions of* linear *ODEs with exponentially decaying forcing, the spectral content is controlled by the forcing spectrum, which itself decays. A rigorous proof of $C_{h_K^*} = O(\rho^K)$ for specific operator classes (e.g., $\mathcal{L}[u] = \ddot{u} + \omega^2 u$) requires explicit Fourier analysis of the HAM recurrence and is left to future work. The proposition is stated conditionally on this spectral decay, which is expected to hold for the standard classes of HAM-solvable ODEs.*

Remark 10 (HAM vs. variable transformations)**.** *The variable transformations of Section 9.1 require the practitioner to identify a single structural feature (exponential, polynomial, frequency) and encode it manually in F. The HAM provides a* systematic *procedure: given only the operator $\mathcal{N}$ and an auxiliary linear operator $\mathcal{L}$, it generates a sequence of linear corrections that progressively absorb the complexity of f^*. No manual identification of structural features is required — the method constructs them automatically from the physics. The price is the computation of K terms of the homotopy series, but the reward is an exponential reduction in sample complexity with increasing K.*

10 Interpretation: Structural Inductive Bias and Effective Hypothesis Space Reduction

10.1 Structural inductive bias

In the explicit formulation $y = \mathrm{MLP}(x;\theta)$, the network must discover *all* structure of f^* from data. The only inductive bias is architectural (depth, width, activation).

In the implicit formulation $F(x, y, \mathrm{MLP}(x;\theta)) = 0$, the function F acts as an *external structural inductive bias*: it encodes prior knowledge about the relationship between x and y that does not need to be learned. This knowledge may come from physical laws, geometric constraints, symmetries, or analytical decompositions (such as the HAM series).

The distinction is analogous to the difference between a noninformative and an informative prior in Bayesian inference: the informative prior (when correct) concentrates the posterior more rapidly, requiring fewer observations. In our framework, the "prior" is the structure of F, and the "concentration of the posterior" manifests as reduction of the Rademacher complexity.

10.2 Effective hypothesis space reduction

The nominal hypothesis class $\mathcal{H}_B$ is the same in both formulations (same network, same weights). What changes is the *effective hypothesis space*: the subset of $\mathcal{H}_B$ relevant to the problem.

In the explicit formulation, the network must cover all functions ε_0-close to f^*:

$$\mathcal{H}^{\mathrm{eff}}_{\mathrm{exp}} = \{g \in \mathcal{H}_{B_{\mathrm{exp}}} : \sup_x |g(x) - f^*(x)| \le \varepsilon_0\}. \tag{37}$$

In the implicit formulation, the network only needs to cover functions ε_0-close to h^*, which is simpler:

$$\mathcal{H}^{\mathrm{eff}}_{\mathrm{imp}} = \{g \in \mathcal{H}_{B_{\mathrm{imp}}} : \sup_x |g(x) - h^*(x)| \le \varepsilon_0\}. \tag{38}$$

Since $B_{\mathrm{imp}} < B_{\mathrm{exp}}$, the effective implicit space has lower "volume" in the sense of Rademacher complexity. The network retains its expressive capacity, but the implicit model *restricts the search* to a smaller region of parameter space.

Remark 11 (Computational Occam principle)**.** *The implicit formulation implements a version of Occam's principle: among all networks satisfying $F(x, y, \mathrm{MLP}(x;\theta)) = 0$, those with smaller path norm B generalize better. The function F acts as a filter that eliminates hypotheses incompatible with the known structure, reducing effective complexity without reducing the network's expressive capacity.*

Remark 12 (Connection to PAC-Bayes)**.** *The effective space reduction argument is compatible with the PAC-Bayes framework [18], where the generalization bound depends on the KL divergence between posterior and prior distributions over θ. The function F acts as a deterministic constraint that concentrates the posterior mass on $\Theta^*_{\mathrm{imp}} \subset \Theta$, reducing the effective KL divergence. Note that F is not a prior in the strict sense (it is not a probability distribution over θ), but a deterministic constraint that reduces the effective support. The formalization of this connection within PAC-Bayes is left as future work.*

10.3 Why PINNs do not fit this framework

It is instructive to examine why the PINN approach [9] does *not* constitute an instance of the present framework, despite sharing the goal of injecting physical knowledge into neural models.

In a standard PINN, the network $u_\theta(x)$ approximates the PDE solution f^* *directly*, and the loss function includes both a data-fitting term and a PDE residual: $\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda\,\mathcal{L}_{\text{PDE}}$. The PDE term constrains the optimization landscape (it penalizes networks incompatible with the governing equation), but the learning target remains f^* itself. In the notation of this paper: $h^* = f^*$, so $B_{\text{imp}} = B_{\text{exp}}$ and $\mathcal{S}(F) = 1$.

The PINN mechanism is *regularization*: it reduces the effective search space during optimization by penalizing physically inconsistent solutions. The mechanism formalized in Theorem 6 is *target transformation*: it changes what the network must learn, replacing f^* with a genuinely simpler function h^*.

This distinction has a precise consequence. PINN-type regularization may improve optimization convergence and reduce overfitting, but it does not change the Rademacher complexity of the hypothesis class (the class of functions the network *can* represent is unchanged). Target transformation reduces the Rademacher complexity because the approximation target itself is simpler (lower Barron norm, lower path norm).

The two mechanisms are complementary: one could use a target transformation (absorbing known structure in F) together with a PINN-style residual loss on the transformed problem, obtaining the benefits of both.

11 Discussion and Future Directions

11.1 What this paper proves

Theorem 6 establishes that, conditional on Assumption 3, the implicit model $F(x, y, \text{MLP}(x;\theta)) = 0$ achieves a tighter generalization bound than the explicit model $y = \text{MLP}(x;\theta)$. Corollary 7 translates this into an explicit sample complexity ratio. The gain condition (25) is a verifiable criterion that balances the path norm reduction against the Lipschitz amplification introduced by F.

The result is constructive: the hypotheses are stated explicitly, the conditions are verifiable for specific choices of F, and the counterexamples show that the result is tight (not every implicit model helps).

11.2 What this paper does not prove

The paper does not resolve two important questions:

(i) When does Assumption 3 hold? The sufficient conditions of Section 7 cover two important classes (structural factorization and nonlinearity absorption), but do not exhaust all possibilities. A general characterization of which functions F yield $B_{\text{imp}} < B_{\text{exp}}$ for a given class of target functions would require a deeper analysis of the interaction between the spectral structure of f^* and the transformation induced by F.

(ii) Why do certain activation functions outperform others for specific dynamical classes? The framework shows that structural knowledge in F reduces sample complexity, but does not explain at a deeper level why $\sin(\omega t)$ is the "right" activation for oscillatory systems or $e^{-\lambda t}$ for damped systems — beyond the observation that they lead to lower B_{imp}.

11.3 Toward autosimilar function spaces

Both questions point toward a topological framework that we outline here and defer to a companion paper.

The key observation is that the history of function approximation can be read as a sequence of discoveries of *autosimilar bases*, each adapted to a different class of natural structure: Fourier series for oscillatory dynamics, Laguerre functions for damped systems, Chebyshev polynomials for compact-domain approximation, Gaussian RBFs for spatially localized problems, and wavelet bases for multi-scale phenomena.

Each of these bases is the orbit of a single *primitive function* under a family of contractive transformations — precisely the structure of an iterated function system (IFS) in the sense of Hutchinson [15]. The Homotopy Analysis Method [13, 14] provides the missing link: it constructs the solution of a nonlinear ODE as the limit of an orbit $\{v_k = \Gamma_\hbar^k[u_0]\}$ generated by a contractive operator, where the primitive u_0 is the homogeneous solution of the auxiliary linear operator $\mathcal{L}$.

Three consequences follow:

1. The exact solution u^* is the attractor of the IFS, not an external target. Approximation is an intrinsic property of the space the solution inhabits, not an operation imposed from outside.

2. Every HAM scheme defines a neural architecture whose activation function is u_0 and whose basis is generated by the HAM contraction. Conversely, every neural architecture whose activation is the homogeneous solution of some $\mathcal{L}$ is an instance of a HAM scheme.

3. The canonical neural architectures (SIREN, RBF, Chebyshev, wavelet networks) are the neural counterparts of the canonical HAM operators.

These claims require precise definitions of autosimilar function spaces, contractivity conditions on $\Gamma_\hbar$, and a formal bridge theorem connecting HAM bases to neural architectures. Their development and proof constitute the subject of a companion paper [19], in which the topological framework provides the structural explanation for *why* Assumption 3 holds for physics-informed architectures, complementing the statistical guarantee established here.

12 Summary of Results

Result	Equation
Explicit generalization bound	$O\left(\frac{M_{\text{exp}}\, B_{\text{exp}}\, X_{\max}\sqrt{L \log d}}{\sqrt{N}}\right)$
Implicit generalization bound	$O\left(\frac{\beta\, M_{\text{imp}}\, B_{\text{imp}}\, X_{\max}\sqrt{L \log d}}{\sqrt{N}}\right)$
Sample complexity ratio	$\frac{\beta^2 M_{\text{imp}}^2 B_{\text{imp}}^2}{M_{\text{exp}}^2 B_{\text{exp}}^2}$
Structural content $\mathcal{S}(F)$	$B_{\text{exp}}/B_{\text{imp}}$
Gain condition	$\mathcal{S}(F) > \beta \cdot M_{\text{imp}}/M_{\text{exp}}$

13 Conclusions

We have established that implicit neural models with structural inductive bias achieve tighter generalization bounds than explicit models, under quantitative conditions that balance the path norm reduction against the Lipschitz amplification introduced by the implicit function F.

The essential mechanism is *target transformation*: the implicit function F changes the learning target from the original f^* to a simpler residual h^*, reducing the intrinsic approximation complexity. This mechanism is fundamentally different from the regularization approach of PINNs, which constrains the optimization landscape but does not change the learning target.

Two instantiations have been identified:

1. **Physics-informed variable transformations**, where known structural features (exponential envelopes, polynomial factors, carrier frequencies) are absorbed by F with $\beta = 1$ and maximal structural content.

2. **Homotopy Analysis Method decompositions**, where the HAM series provides a systematic decomposition of the target function into linear corrections of geometrically decreasing complexity, requiring no manual identification of structural features.

The practical consequence is a principled criterion for evaluating whether a given implicit formulation will outperform the explicit baseline: compute the structural content $\mathcal{S}(F)$ and verify the gain condition $\mathcal{S}(F) > \beta \cdot M_{\text{imp}}/M_{\text{exp}}$.

The deeper question — *why* certain bases are intrinsically adapted to certain dynamics — requires a topological framework based on autosimilar function spaces and the bridge between HAM and neural architectures, to be developed in a companion paper.

Declaration of competing interest

The author declares no competing interests.

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